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Campus Lost & Found Portal: A Java-Based Item Recovery and Management System

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA0908 – Programming in JAVA

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE ENGINEERING
(AI DS)**

Submitted by

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Under the Supervision of

Dr. Meganathan E

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SEPTEMBER 2026



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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Campus Lost & Found Portal” has been carried out by Vedvyass M (192324267) under the supervision of Dr. E. MEGANATHAN and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science Engineering (AI DS) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, Vedvyass M (192324267) of the Artificial Intelligence and Data Science, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Campus Lost & Found Portal: A Java-Based Item Recovery and Management System” is the result of our project work based on the submitted project material.

To the best of our knowledge, the work presented herein represents the system, modules, techniques, workflows, and implementation evidence described in the project presentation. The report preserves the terminology and functionality provided in the source material. Any future enhancements mentioned in the project are presented as proposed extensions rather than completed features.

Place : Chennai

Date : 03/09/2026

Signature of the Students with Names



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ABSTRACT

The Campus Lost & Found Portal is a Java-based application designed to provide a centralized solution for managing lost and found belongings within a college campus. The system allows students and staff to report lost items, register found items, search and filter reported items, and track the status of reports. Administrators can verify reports, manage users, and monitor item recovery. The proposed portal reduces dependency on physical notice boards, informal messaging groups, and manual searching.

The system is organized into three major modules. Module 1 provides user and authentication management, including registration, login/logout, profile management, role-based access, password management, and access control. Module 2, the focus of the individual contribution in this report, manages lost and found item records, image uploads, search and filtering, report updates and deletion, item status management, and administrator review. Module 3 handles item matching, claim submission, ownership verification, claim decisions, recovery, and history.

The implementation material demonstrates Java packages, inheritance through servlets extending HttpServlet, object creation through constructors, database-oriented processing, and structured application workflows. The architecture connects the presentation layer, Java application layer, database layer, administrator functions, and external notification concepts. The resulting system provides a structured, searchable, and maintainable approach to campus item recovery.

Keywords: Java, Lost & Found, Item Management, Servlets, JDBC, MySQL, Authentication, Item Matching

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The Campus Lost & Found Portal is a Java-based application intended to manage lost and found items within a college campus. Students and staff can report lost belongings and register items that they have found. Each item record can contain identifying information such as item name, category, location, date, description, colour, and contact information. The system also supports image uploading to make identification easier.

The project is designed as a centralized alternative to informal reporting through WhatsApp groups, notice boards, and manual searching. The system provides a structured flow from item reporting to search, status management, administrator review, and, through Module 3, matching and recovery.

1.2 Engineering Context

The engineering problem is not limited to storing an item record. The application must provide controlled access, organized item data, searchable reports, editable submissions, status tracking, and administrative verification. The system architecture therefore separates the presentation layer, Java application layer, database layer, administrator functions, and external service concepts.

1.3 Problem Statement

Lost items on campus are often reported through informal channels such as WhatsApp groups or notice boards. There is no centralized system to record and track lost and found belongings. Students may find it difficult to identify whether their lost item has been recovered, while manual management can create duplicate, outdated, or incomplete information.

The proposed engineering solution is a centralized, searchable, and secure platform that records lost and found reports and supports the recovery process. The portal provides a single place for users to submit item information and for administrators to review and manage reports.

Problem	Effect	System Response
Informal reporting channels	Reports are scattered and difficult to track	Centralized item reporting
No searchable record	Users spend time manually searching	Search and filter functionality
Incomplete item information	Identification becomes difficult	Structured item fields
Outdated report status	Users cannot know the current state	Lost / Found / Claimed / Returned status
Unverified listings	Incorrect or inappropriate reports may remain	Administrator review and verification

1.4 Objectives and Scope

Project Objectives

1. Develop a centralized portal for reporting lost and found items.
2. Allow users to register, search, and track items easily.
3. Provide item matching based on category, location, date, and description.
4. Allow administrators to verify and manage submitted reports.
5. Maintain a database of lost and found items.
6. Improve the chances of returning lost belongings to their rightful owners.
7. Reduce manual effort and communication gaps.

Scope

- Lost-item and found-item reporting.
- Structured item details including name, category, colour, description, date, and location.
- Image upload for easier identification.
- Search and filtering by category, location, date, and keywords.
- Update and deletion of user-submitted reports.
- Item status management.
- Administrator review and verification.
- Integration with the authentication and claim/recovery modules.

1.5 Expected Outcomes and Project Benefits

The expected outcome is a centralized campus item-recovery system that makes reporting and searching faster and easier. The portal is intended to improve organization of item information and reduce communication gaps between people who lose items, people who find items, and administrators.

Outcome	Benefit
Centralized records	One structured location for lost and found reports
Image support	Easier visual identification of an item
Search and filter	Faster discovery of potentially relevant reports
Status tracking	Clearer visibility of Lost, Found, Claimed, and Returned states
Admin verification	Improved control over reported listings
Module integration	Connection between reporting, matching, claiming, and recovery

The project presentation also identifies future enhancement possibilities including AI, mobile applications, notifications, and cloud technologies. These are considered future extensions rather than completed capabilities in the submitted material.

CHAPTER 2

SYSTEM ANALYSIS AND EXISTING APPROACH

2.1 Existing Approach

The project identifies informal reporting through WhatsApp groups and notice boards as the existing practical approach. Such channels can communicate a lost-item message quickly, but they do not provide a structured database, controlled status lifecycle, centralized search, or systematic administration of reports.

2.2 Proposed Approach

The proposed portal replaces fragmented reporting with a Java-based application and database-backed workflow. Users interact through the presentation layer, Java application components process requests, and the database layer stores the structured records. Administrator functions provide verification and management.

Aspect	Existing Informal Approach	Proposed Portal
Reporting	Messages / notice boards	Structured item forms
Storage	Scattered messages	Database records
Search	Manual	Category, location, date, keyword filters
Status	Not consistently maintained	Lost, Found, Claimed, Returned
Verification	Informal	Administrator review
Recovery workflow	Communication dependent	Connected to matching and claims module

2.3 System Architecture / Workflow

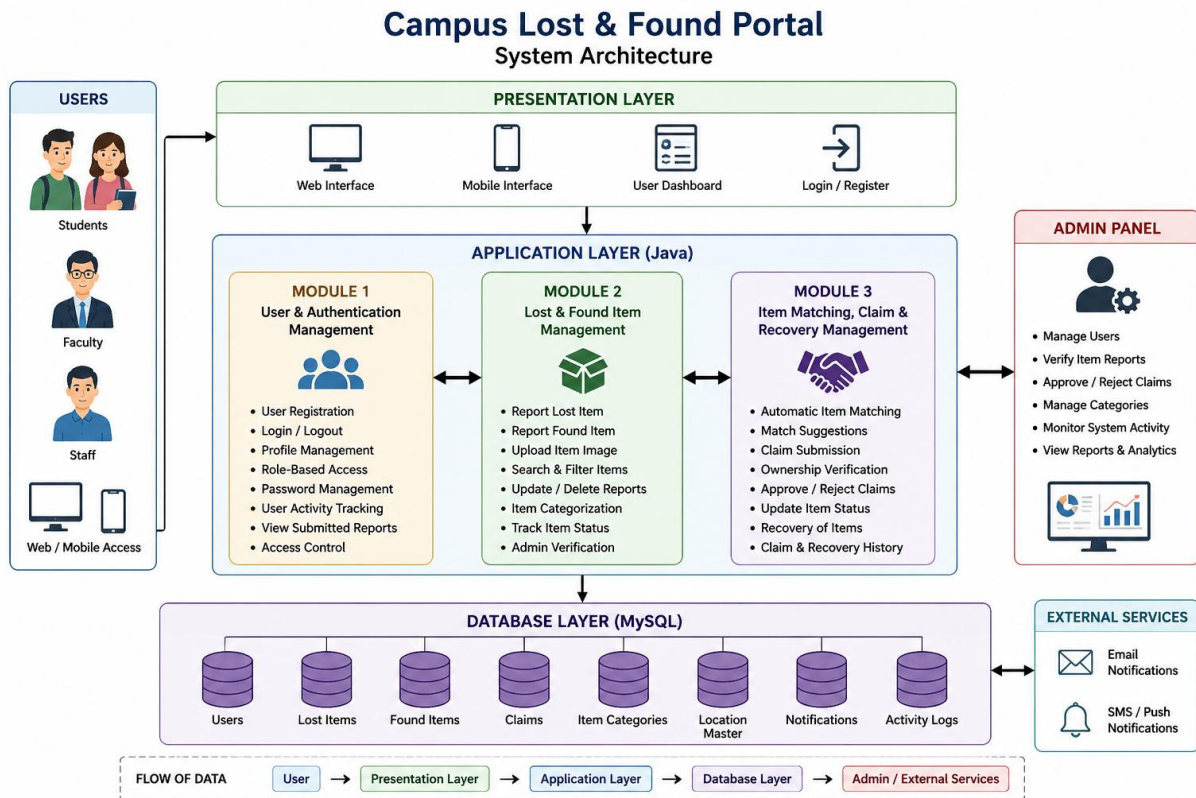


Figure 2.1: Campus Lost & Found Portal – System Architecture

The architecture shown in the project presentation separates the system into a presentation layer, application layer, and database layer. The application layer contains Module 1, Module 2, and Module 3. The architecture also shows an administrator panel and external services for notification concepts. Module 2 sits at the centre of item information management and connects reporting activity with the database and the subsequent matching and recovery workflow.

2.4 Module Overview

Module	Purpose	Key Functions
Module 1	User & Authentication Management	Registration, login/logout, profiles, role-based access, password management, submitted reports, access control
Module 2	Lost & Found Item Management	Report lost/found items, image upload, search/filter, update/delete, status management, admin verification
Module 3	Item Matching, Claim & Recovery Management	Matching, claim submission, ownership verification, approve/reject claims, recovery, history

2.5 Relationship Between Modules

Module 2 provides the item records that are later used by Module 3 for matching and recovery. Module 1 controls the user identity and access context under which reports are submitted and managed. This modular structure allows each major responsibility to be separated while keeping the complete portal workflow connected.

2.6 Data and Item Record Structure

The project presentation identifies the MySQL database layer and an items table for Module 2. The item-management workflow is organized around the stored item record and the actions performed on it.

Item Information	Purpose in Module 2
Item name	Identifies the reported belonging
Category	Supports classification and filtering
Colour	Provides an additional identifying characteristic
Description	Stores detailed identifying information
Date	Supports time-based search and identification
Location	Identifies where the item was lost or found
Image	Provides visual identification support
Status	Tracks Lost, Found, Claimed, and Returned states



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The source presentation also identifies contact information as part of the broader reporting context. The exact database column definitions are not provided in the presentation; therefore this report does not invent a detailed schema beyond the stated item-management fields.

CHAPTER 3

ENGINEERING DESIGN, REQUIREMENTS, SECURITY AND RISK

3.1 Stakeholder Requirements

Stakeholder	Requirement
Students	Report lost/found items and search existing reports
Faculty / Staff	Register found items and search for relevant belongings
Administrators	Review, verify, and manage item reports
System / Developer	Maintain structured Java application components and database processing
Project Users	Receive a simple workflow for reporting and finding items

3.2 Functional and Non-Functional Requirements

Requirement Type	Requirement
Functional	Create lost and found item reports
Functional	Upload an item image
Functional	Search and filter item records
Functional	Update or delete submitted reports
Functional	Maintain item status
Functional	Administrator review and verification
Functional	Store item information in the database
Non-Functional	Usability: simple reporting and searching
Non-Functional	Maintainability: organized packages and module separation
Non-Functional	Access control: restricted administrative functions
Non-Functional	Reliability: validate item information and handle database errors

3.3 Design Specifications

Parameter	Target / Requirement	Priority
Input	Lost/found item details	High
Database	items table for item records	High
Image handling	Upload item image	High
Search	Category, location, date, keywords	High
Editing	Update/delete own reports	High
Status	Lost, Found, Claimed, Returned	High
Verification	Administrator review	High
Java structure	Packages and servlet-based inheritance	High

3.4 Module 2 Design Principle

Module 2 is designed as the item-management centre of the portal. Its responsibility begins with capturing item information and continues through storage, image support, retrieval, modification, deletion, status tracking, and administrator verification. The module therefore acts as the operational bridge between users and the item database.

3.5 Module 2 Feature Specification

Feature	Input / Action	Expected Result
Report Lost Item	User enters item details	Lost item record is created
Report Found Item	User enters found-item details	Found item record is created
Image Upload	User selects an item image	Image is associated with item identification
Search & Filter	Category / location / date / keyword	Relevant item records are displayed
Update Report	User edits submitted report	Updated item information is stored
Delete Report	User removes own report	Selected report is removed
Status Management	Status is maintained by workflow	Item reflects current state
Admin Review	Administrator checks report	Report is verified or managed

3.6 Design Trade-offs

Design Decision	Alternative	Benefit	Trade-off
Centralized database	Informal messages	Structured and searchable records	Requires database management
Image upload	Text-only reports	Better item identification	Requires image handling
Search and filters	Manual browsing	Faster discovery	Requires consistent item fields
Status lifecycle	No status tracking	Clearer recovery state	Status must be maintained correctly
Admin verification	Open publishing	Better control over listings	Requires administrator effort
Modular Java design	Single large component	Improves organization	Requires coordination between modules

3.7 Security, Safety and Risk

The project is a software-based campus management application and does not directly control physical equipment. The principal safety concern is therefore the effect of incorrect or misleading information rather than physical system control. Access control, administrator verification, and structured records are important for reducing misuse.

Risk	Impact	Mitigation
Incorrect item information	Difficult identification	Validate and review submitted reports
Duplicate or outdated reports	Confusion during searching	Maintain report status and update/delete capability
Unauthorized administrative action	Incorrect listings or decisions	Role-based access from Module 1
Database operation failure	Report may not be stored correctly	Structured Java database handling and exception handling
Inappropriate image/report	Poor quality or misuse of portal	Administrator review

The source presentation demonstrates exception handling using try-with-resources and SQLException handling. This supports controlled handling of database operations and helps prevent unhandled database errors.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project is implemented as a Java-based application with a database-backed workflow. The presentation identifies Java, servlets, packages, JDBC-related database connectivity, MySQL database storage, and Java programming concepts such as inheritance and constructors. The complete system is divided into three functional modules.

Technology / Concept	Purpose
Java	Application logic and implementation
Servlets / HttpServlet	Receive and process application requests
JDBC	Database connectivity and database operations
MySQL	Storage layer identified in the architecture
Packages	Organized code structure using com.lostfound.*
Inheritance	Servlets extending HttpServlet
Constructors	Object creation and initialization
Exception Handling	Controlled database error handling

4.2 Module 2 Implementation Evidence

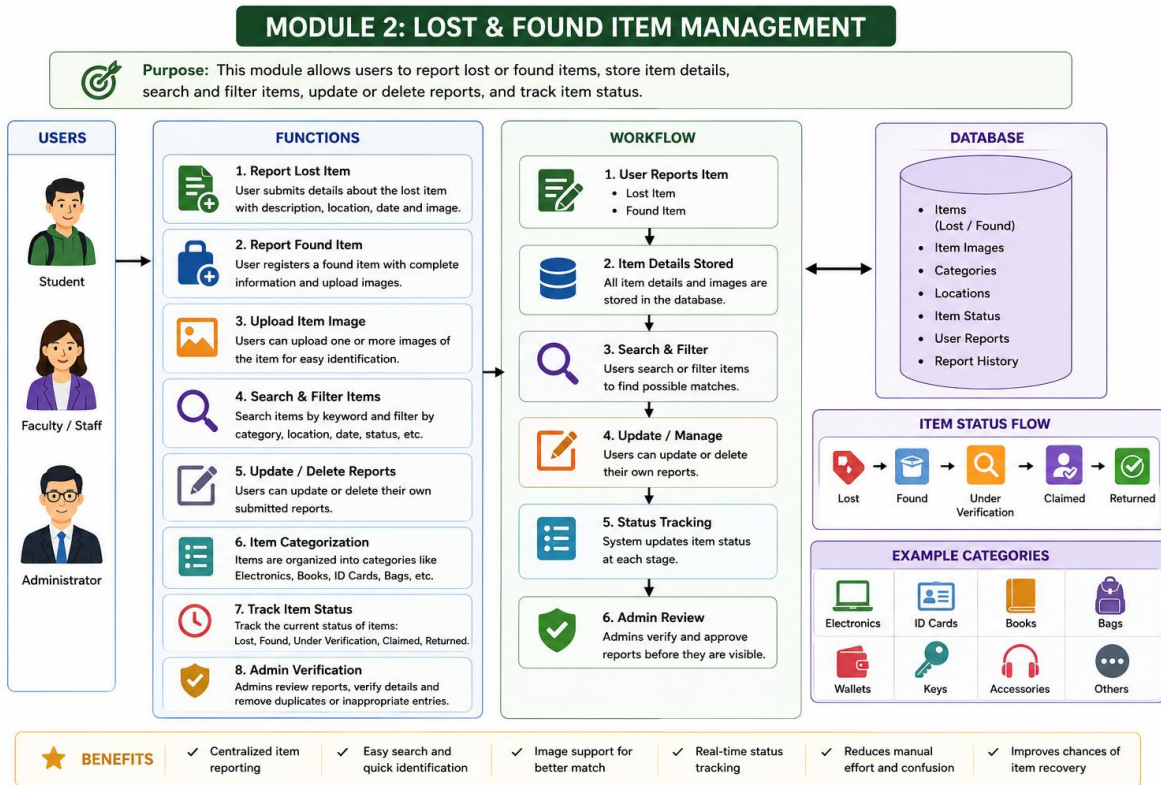
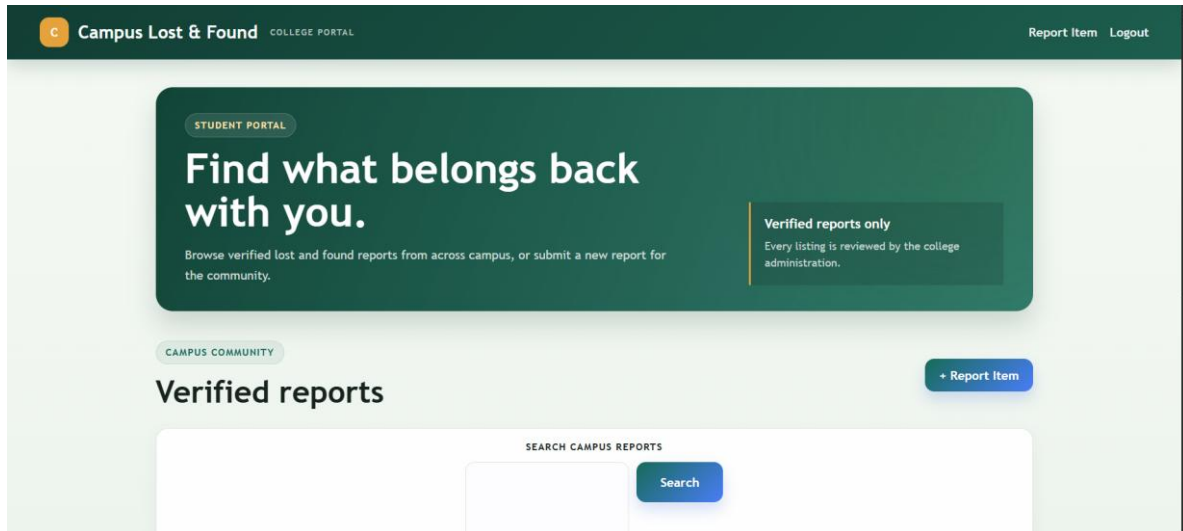


Figure 4.1: Module 2 – Lost & Found Item Management

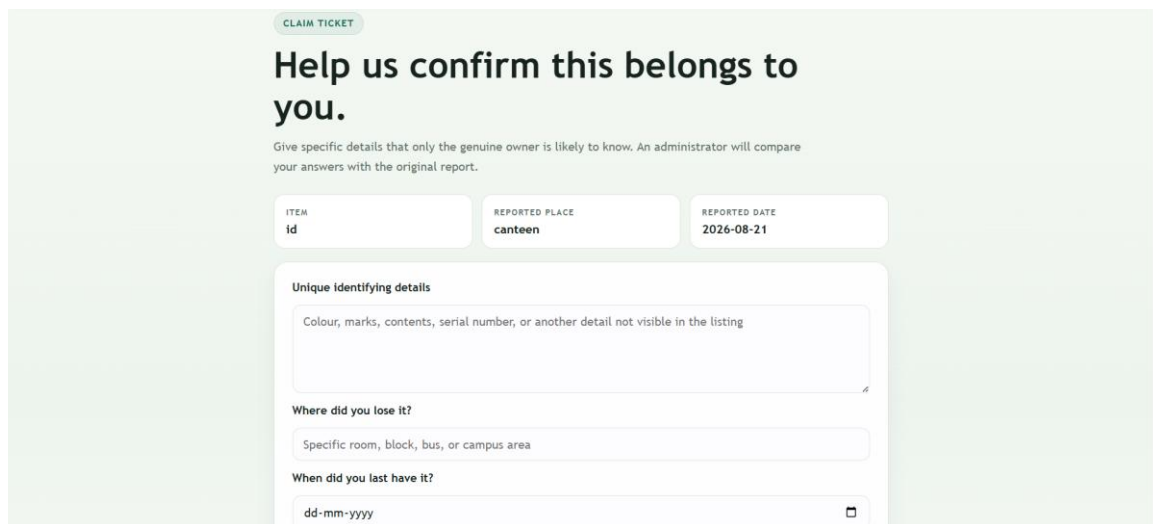
The Module 2 architecture shown in the project presentation identifies the user-facing functions, workflow, database interaction, and expected benefits. The module includes report-lost and report-found functions, image upload, search and filtering, update/delete operations, status tracking, and administrator verification.

4.3 Module 2 User Interface



The screenshot shows the 'Campus Lost & Found' interface. At the top, there's a header with 'Campus Lost & Found' and 'COLLEGE PORTAL'. A 'Report Item' button and a 'Logout' link are in the top right. The main content area has a 'STUDENT PORTAL' section with the heading 'Find what belongs back with you.' and a subtext: 'Browse verified lost and found reports from across campus, or submit a new report for the community.' To the right, a box states 'Verified reports only' and 'Every listing is reviewed by the college administration.' Below this, there's a 'CAMPUS COMMUNITY' section with the heading 'Verified reports' and a '+ Report Item' button. A search bar labeled 'SEARCH CAMPUS REPORTS' with a 'Search' button is at the bottom.

Figure 4.2: Verified reports / item-management interface



The screenshot shows the 'CLAIM TICKET' interface. It starts with the heading 'Help us confirm this belongs to you.' and a subtext: 'Give specific details that only the genuine owner is likely to know. An administrator will compare your answers with the original report.' Below this, there are three input fields: 'ITEM id', 'REPORTED PLACE' (with the value 'canteen'), and 'REPORTED DATE' (with the value '2026-08-21'). A section titled 'Unique identifying details' contains a large text area for 'Colour, marks, contents, serial number, or another detail not visible in the listing'. Below this, there are two more input fields: 'Where did you lose it?' (with the value 'Specific room, block, bus, or campus area') and 'When did you last have it?' (with the value 'dd-mm-yyyy').

Figure 4.3: Item details and reporting interface

4.4 Module 2 Java Techniques

Packages

The source presentation identifies packages as an organized code structure using `com.lostfound.\*`. This keeps related Java components grouped into a maintainable application structure.

Inheritance

Servlets extend the `HttpServlet` class. This demonstrates inheritance by deriving application-specific servlet classes from the servlet framework class.

Object Creation Using Constructor

```
User user = new User(name, email, password, department, phone, "USER");

public class User {
    public User() {} // No-argument constructor

    public User(String name, String email, String password, ...) {
        this.name = name;
        this.email = email;
        // Initialize all fields
    }
}
```

The constructor example demonstrates object creation using `new` and initialization of object fields. Although the example is presented under the Module 2 technique slide, it reflects the Java object-oriented structure used across the portal.

4.5 Module 2 Workflow

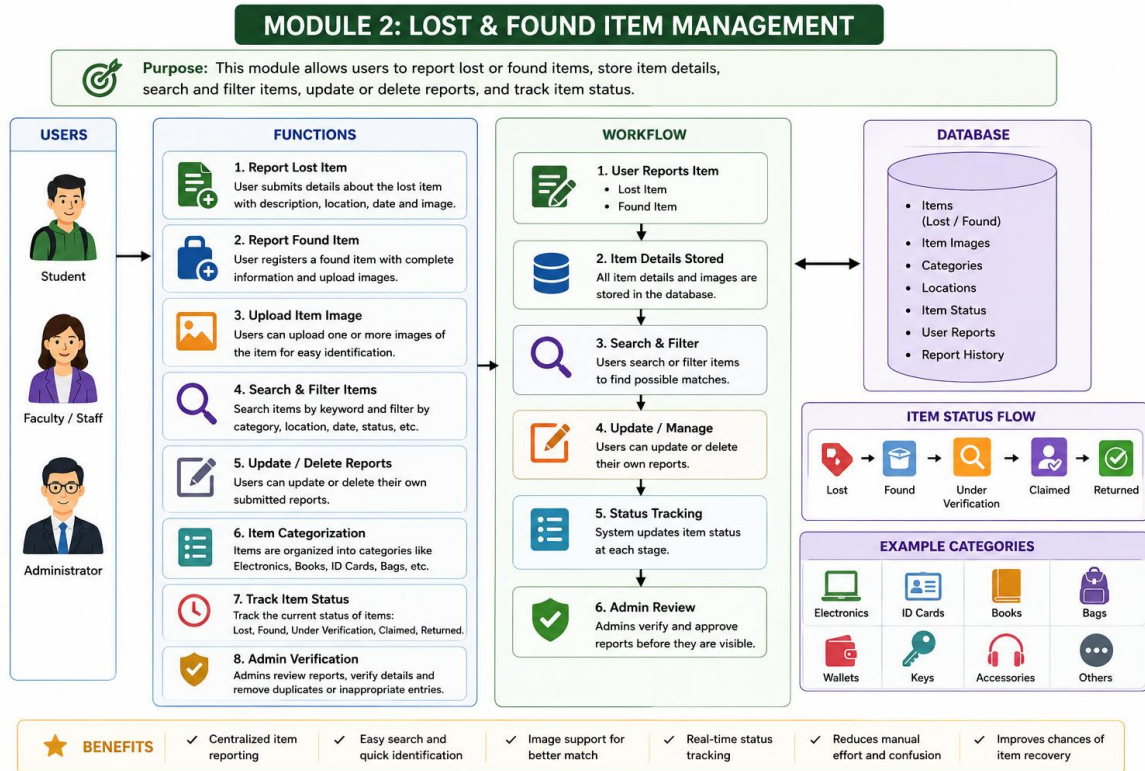


Figure 4.4: Module 2 workflow and database relationship

1. A user reports a lost or found item with the required details.
2. The item information is associated with the MySQL items table.
3. An image can be uploaded to support identification.
4. Search and filtering allow users to locate relevant reports.
5. Users can update or delete their own submitted reports.
6. The item status is maintained as Lost, Found, Claimed, or Returned.
7. Administrators review and verify reported items before the information is relied upon in the recovery process.

4.6 Test Plan & Verification

Test Case	Test Method	Expected Result
Report lost item	Submit a complete lost-item form	Lost item record is created
Report found item	Submit a complete found-item form	Found item record is created
Image upload	Attach an item image	Image is accepted for identification
Search	Search using a keyword	Relevant reports are displayed
Filter	Apply category/location/date filter	Matching records are narrowed
Update	Edit an existing user report	Changed details are stored
Delete	Delete an own report	Selected report is removed
Status	Change/track item state	Valid status is maintained
Admin review	Administrator checks a report	Report can be verified or managed

The source presentation does not provide numerical test results or formal pass/fail measurements. Therefore, this report records the verification criteria supported by the stated module functions without inventing performance statistics.

4.7 Individual Contribution – Module 2

Individual contributor: Vedvyass M (192324267)

The individual contribution for this capstone is Module 2 – Lost & Found Item Management. The module is responsible for the core item-record workflow: reporting lost items, registering found items, storing identifying information, uploading images, searching and filtering records, updating or deleting submitted reports, maintaining item status, and supporting administrator review.

Contribution Area	Work Represented in the Project
Item reporting	Lost and found item entry workflow
Item data	Name, category, colour, description, date, location and related details
Image support	Item image upload for identification
Search / filter	Category, location, date and keyword-based discovery
Report maintenance	Update and delete operations for submitted reports
Status management	Lost, Found, Claimed, Returned lifecycle

Admin support	Review and verification of item listings
Java concepts	Packages, inheritance, constructors and organized module structure

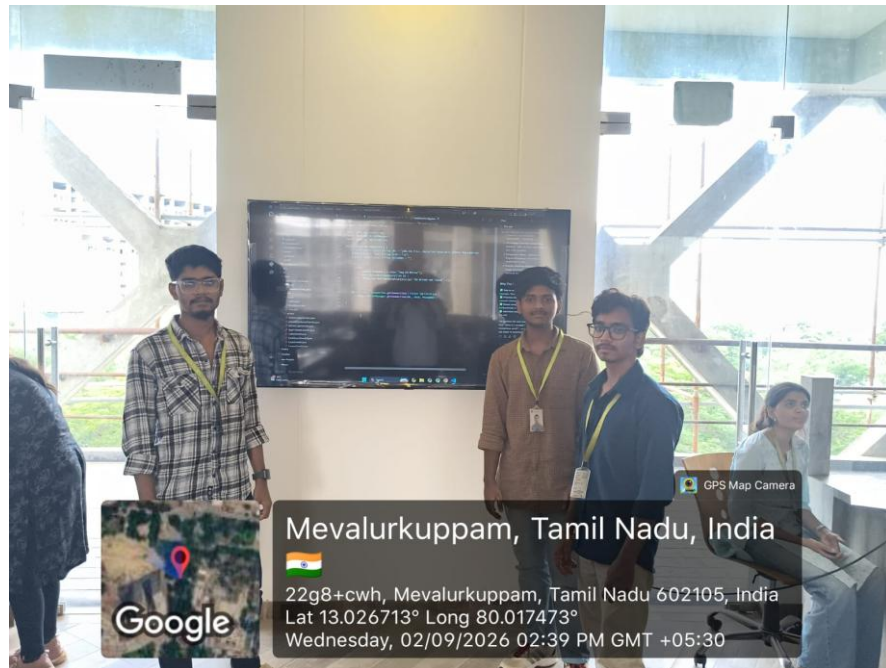


Figure 4.5: Team Project Evidence

CHAPTER 5

CONCLUSION & REFLECTION

The Campus Lost & Found Portal provides a centralized solution for managing lost and found belongings within a college campus. It makes reporting and searching for items faster and easier by replacing fragmented communication with structured item records and a searchable workflow.

Module 2 forms a critical part of the system because it manages the item lifecycle from initial reporting through search, modification, deletion, status tracking, and administrator verification. Its integration with Module 1 provides user and access context, while Module 3 uses the item information for matching, claim, and recovery operations.

The project demonstrates how Java can be used to organize application logic through packages, inheritance, constructors, servlets, and database-oriented processing. The system can be further enhanced with AI, mobile applications, notifications, and cloud technologies as identified in the project presentation.

From an engineering perspective, the project highlights the importance of centralized data management, modular design, controlled access, structured item information, and verification. The individual Module 2 contribution demonstrates how a focused module can support the overall system while remaining connected to the other functional components.



REFERENCES

The supplied project presentation was used as the primary source for the content of this final report. No external reference list was provided in the presentation; therefore, no external sources have been added or represented as if they were part of the submitted project material.

Reference	Use in Report
CSA0908 – Java PPT Final, Campus Lost & Found Portal: A Java-Based Item Recovery and Management System	Primary project source for objectives, modules, architecture, techniques, interfaces and conclusion
Project implementation examples shown in the supplied presentation	Source for the Java package, inheritance and constructor examples reproduced in the implementation discussion

APPENDICES

Appendix A – Module 2 Technique Extract

Packages - Organized code structure (`com.lostfound.\*`)

Inheritance - Servlets extending `HttpServlet` class.

Appendix B – Constructor Example

```
User user = new User(name, email, password, department, phone, "USER");

public class User {
    public User() {}
    public User(String name, String email, String password, ...) {
        this.name = name;
        this.email = email;
        // Initialize all fields
    }
}
```

Appendix C – Module 2 Feature Flow

MySQL → items table → Image Upload → Search & Filter → Update / Delete Report → Item Status Management → Admin Review & Verification.

Appendix D – Source Presentation Screens

The report incorporates the architecture and interface evidence supplied in the project presentation, including the system architecture, Module 2 design, verified-report interface, item-detail interface, and team project evidence.